KABALE UNIVERSITY

P. O. Box 317, Kabale - Uganda
Plot 364 Block 3 Kikungiri Hill
Kabale Municipality

FACULTY OF COMPUTING LIBRARY AND INFORMATION SCIENCE

DEPARTMENT OF COMPUTER SCIENCE

BCS 3101 BASICS OF MACHINE LEARNING

Capstone Project Report

Diabetes Risk Prediction from Routine Clinical Measurements

An End-to-End Machine Learning Solution for a Real-World Problem

Project team

NAME	REG.No
KADUUMA COSMAS ALLAN	2024/A/KCS/4930/F
MUHOOZI TRACY	2024/A/KCS/3103/G/F
MUHEREZA ALOUZIOUS	2024/A/KCS/5193/G/F
AMUTUHAIRE TRACY	2024/A/KCS/5192/G/F
WAKHABEKO EMMA	2024/A/KCS/3104/G/F
AHURIRA LUCKY	2024/A/KCS/5197/G/F
MUGISHA ALEX	2024/A/KCS/5461/F

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1. Introduction and Background

Diabetes is a long-term condition in which the body cannot keep blood sugar at a safe level. Over the years it can damage the eyes, kidneys, nerves and heart. The hard part is that many people live with high blood sugar for a long time without knowing, because early diabetes often shows no clear signs.

The sure way to confirm diabetes is a blood glucose test. In a busy hospital or a small health centre, that test costs money and time and a nurse often has to decide who gets looked at first. A tool that can point to the patients most likely to have diabetes would help with that decision.

This project asks a simple question: can a few ordinary clinical measurements, such as glucose reading, body mass index (BMI), blood pressure, age and number of pregnancies, be used to predict whether a woman has diabetes? To answer it, we trained and compared several machine learning models on a well known public dataset.

We tried to follow the full path of a real project. We defined the problem, read what others have done, prepared the data carefully, compared seven models fairly, tuned the best ones, looked closely at the mistakes and saved the final model so that it can be reused outside the notebook.

2. Literature Review

Before building anything we looked at six sources: three on diabetes prediction, one on the dataset itself and two on the method and the tools. Here is what each one contributes.

2.1 Smith et al. (1988): the origin of the dataset

Smith and colleagues used the ADAP learning algorithm to forecast who would develop diabetes in a group of Pima Indian women. This is the study that gave us the dataset used here. Their algorithm gave a score between 0 and 1, which they turned into a yes or no using a cutoff of 0.448. With 576 training cases, it reached about 76% sensitivity and specificity on the remaining 192.

This gave us a first yardstick. A modern model on the same data should reach roughly the same level and a score of about 76% is a fair target, not a failure.

2.2 Kavakiotis et al. (2017): a review of the field

This paper is a systematic review of machine learning in diabetes research. It found that about 85% of the methods used were supervised learning and that support vector machines were the most popular and most successful. It also noted that most work used clinical datasets.

This supports our decision to treat the problem as supervised classification and to include an SVM among the models.

2.3 Zou et al. (2018): trees, forests and neural networks

Zou and co-authors predicted diabetes using hospital check-up data from Luzhou, China, with decision tree, random forest and neural network models. They used five-fold cross validation

and tried PCA and mRMR to cut down the number of features. Random forest did best, with an accuracy of 0.8084 when all attributes were used.

From this we took the idea of testing tree-based models and of validating with cross validation instead of a single split. It also suggested PCA as a tool worth trying.

2.4 Maniruzzaman et al. (2017): comparing classifiers

This paper compared several classifiers for diabetes data, including Gaussian process classification against linear discriminant analysis, quadratic discriminant analysis and naive Bayes, using cross validation. The Gaussian process model gave the best accuracy, sensitivity and specificity in their experiments.

It reminded us that a simple probabilistic model like naive Bayes is a fair candidate to include and that sensitivity (finding the sick patients) should be reported next to accuracy.

2.5 Kaufman et al. (2012): data leakage

Kaufman and co-authors describe leakage as information about the target that should not be available when the model is built. They call it one of the top ten data mining mistakes. In practice, a very common form is fitting a scaler or imputer on the whole dataset before splitting it.

This is why our pipeline splits the data first and fits every preparation step on the training part only.

2.6 Pedregosa et al. (2011): scikit-learn

This is the paper behind scikit-learn, the Python library we used for all the models, the pipeline, cross validation and grid search. It provides tested implementations of every algorithm in this report, which makes the work easy to repeat.

2.7 What this means for our choices

Putting these together, a spread of linear, distance-based, probabilistic, tree-based and kernel models is a reasonable start, not a random one. The literature suggests that results on this kind of data sit around the mid-to-high 70s for accuracy, that a few models are usually close together and that honest validation matters more than picking a fancy algorithm.

3. Problem Statement and Objectives

3.1 Problem statement

Health workers in district hospitals and health centres, including those in South Western Uganda, often see more patients than they can fully test. Diabetes can stay hidden for years and by the time it is found, damage may already be done. Someone has to decide which patients need a full diagnostic check and follow-up first.

The people who care about the answer are the nurses and clinicians who make that call, the patients who depend on being picked up early and the health managers who have limited tests

to spend. The decision this project informs is simple: should this patient be flagged for a confirmatory test and follow-up?

We do not have patient data from Uganda, so we used a public dataset to build and test the method. This matters and we come back to it in the limitations and in the Responsible AI section.

3.2 General objective

To build and evaluate a leakage-safe machine learning pipeline that flags women likely to have diabetes from routine clinical measurements and to show clearly how well it works and where it fails.

3.3 Specific objectives

1. Acquire a suitable public dataset, check its quality and find any hidden missing values.
2. Build a preparation pipeline that fits the imputer and scaler on training data only.
3. Train and compare at least four different models with repeated cross validation, reporting both the mean and the spread of the scores.
4. Tune the two strongest models with GridSearchCV and choose the best one using a clear rule.
5. Test the chosen model once on held-out data, pick a sensible decision threshold and study its errors and subgroup performance.
6. Save the final pipeline with joblib and show it making a prediction after being loaded back.

3.4 Target, features and baseline

Table 1: Task definition

Item	Description
Target variable	Outcome (1 = diabetes, 0 = no diabetes)
Features	Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age
Baseline	Always predict “no diabetes”. This gives 65.1% accuracy in cross validation, but 0% recall and a ROC-AUC of 0.50. Any useful model must beat this on recall and ROC-AUC.

Notice that accuracy alone is a weak test here, because 65% of patients do not have diabetes. A model that never finds a single sick patient would still look “65% accurate”. That is why we focus on recall and ROC-AUC as well.

4. Data and Methodology

4.1 Source, licence and how it was collected

The data is the Pima Indians Diabetes Database. It comes from the National Institute of Diabetes and Digestive and Kidney Diseases in the USA and was donated to the UCI Machine Learning Repository. It is also widely shared on Kaggle, where some copies are listed as public domain (CC0). We used a CSV copy of it with all 768 records. Anyone reusing it should check the licence on the page they download from and cite Smith et al. (1988).

The women in the study live near Phoenix, Arizona and were examined regularly. Diabetes was recorded using World Health Organization criteria, which include a 2-hour glucose reading above 200 mg/dl.

4.2 What is in the data

There are 768 patients and 8 features. Of these, 268 (34.9%) have diabetes and 500 (65.1%) do not. All patients are women aged 21 or older.

Table 2: The eight features

Feature	Meaning
Pregnancies	Number of times pregnant
Glucose	Plasma glucose 2 hours into an oral glucose tolerance test
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skin fold thickness (mm)
Insulin	2-hour serum insulin (mu U/ml)
BMI	Body mass index (kg/m^2)
DiabetesPedigreeFunction	A score of diabetes in the family history
Age	Age in years

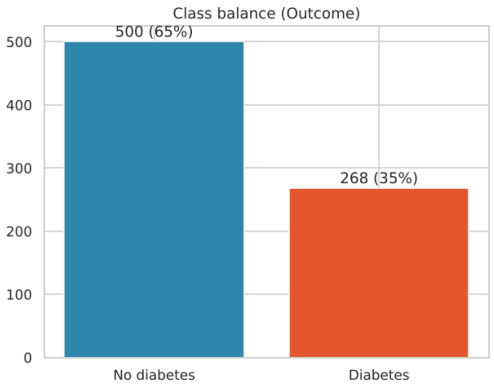


Figure 1: Class balance. About one in three patients has diabetes.

4.3 Limitations of the data

First, it is a small dataset from one community, so the results may not carry over to other populations, including Ugandan patients. Second, all the patients are women, so the model says nothing about men.

Third, the data is old and was collected for research, not routine care. Fourth, the diagnosis is based on a glucose test and Glucose is also one of the features. So a strong link between the two is expected and the model is closer to a second opinion than to a true early warning.

4.4 Hidden missing values

At first sight there are no missing values. But some columns contain zeros that cannot be real. A person cannot have a blood pressure or BMI of zero. These zeros are missing values written as 0.

Table 3: Hidden missing values (zeros)

Column	Zeros	Share of rows
Glucose	5	0.7%
BloodPressure	35	4.6%
SkinThickness	227	29.6%
Insulin	374	48.7%
BMI	11	1.4%

We changed these zeros to missing values (NaN). We did not delete the rows, because that would throw away almost half of the data for Insulin. The gaps are filled later, inside the pipeline.

4.5 Key findings from exploring the data

Glucose has the strongest link with diabetes (correlation 0.495), followed by BMI (0.314), Insulin (0.303), SkinThickness (0.259), Age (0.238) and Pregnancies (0.222). Blood pressure and the family history score are weaker (about 0.17 each). The histograms, boxplots and heatmap are in Appendices A to C.

The boxplots show that patients with diabetes tend to have higher glucose, higher BMI and to be older. The two groups overlap a lot though, which tells us no single measurement will separate them cleanly.

4.6 The leakage-safe pipeline

We split the data first: 80% for training (614 patients) and 20% for testing (154 patients). The split is stratified, so both parts keep the same share of diabetes (34.9% in training and 35.1% in testing). The test set was kept locked away and used only once at the end.

Every model sits inside a scikit-learn Pipeline with three steps: fill missing values with the median, scale the features with StandardScaler, then fit the model. Because it is one pipeline, cross validation refits the imputer and scaler on each training fold only. Nothing is learned from the validation fold or the test set.

5. Model Architecture and Design

We trained seven different models plus a baseline. All of them use the same random seed (42) so the results can be repeated. Table 4 lists the settings and the paragraphs after it explain how each model makes a prediction.

Table 4: Models and starting settings

Model	Type	Key settings
Baseline	Dummy classifier	Always predicts the most common class
Logistic Regression	Linear	C = 1.0, L2 penalty, max_iter = 1000
KNN	Distance based	15 neighbours, uniform weights
SVM (RBF)	Kernel	C = 1.0, gamma = “scale”, probability output on
Naive Bayes	Probabilistic	Gaussian, default smoothing
Decision Tree	Tree	max_depth = 4, min_samples_leaf = 5
Random Forest	Ensemble of trees	300 trees, min_samples_leaf = 3
Gradient Boosting	Ensemble of trees	150 trees, learning rate 0.05, max_depth = 2

Baseline. It ignores the features and always says “no diabetes”. It exists only to show what a model must beat.

Logistic Regression. It gives each feature a weight, adds them up and turns the total into a probability between 0 and 1. We kept C at 1.0 (a moderate penalty that stops the weights from growing too large) and allowed up to 1000 iterations so the fit converges. Its weights can be read directly, which makes it easy to explain.

K-Nearest Neighbours. It stores the training patients. For a new patient it finds the 15 most similar ones and takes a vote. We chose 15 so that one odd patient cannot swing the answer. Scaling matters a lot here, which is why the pipeline scales first.

SVM with an RBF kernel. It looks for a boundary that separates the two groups with the widest possible gap and the RBF kernel lets that boundary curve. C controls how strict it is about mistakes and gamma controls how local the curve is. We turned probability output on so we could use thresholds and ROC curves.

Naive Bayes. It works out how likely each feature value is for diabetic and non-diabetic patients, assuming a bell-shaped spread and combines them using Bayes' rule. It is fast and has almost nothing to tune.

Decision Tree. It asks a chain of yes or no questions, such as “is glucose above a certain value?”, until it reaches an answer. We limited the depth to 4 and required at least 5 patients per leaf, because a deep tree just memorises the training data.

Random Forest. It grows 300 trees, each on a random sample of the data and features and lets them vote. Averaging many different trees reduces the over-fitting of a single tree. A leaf size of 3 keeps each tree reasonably simple.

Gradient Boosting. It builds small trees one after another and each new tree tries to fix the mistakes of the ones before it. We used shallow trees (depth 2) and a small learning rate (0.05), which makes it learn slowly and carefully.

6. Results and Analysis

6.1 Comparing all models with cross validation

Every model was scored with 5-fold cross validation repeated 3 times, which gives 15 scores per model. Table 5 shows the mean and the spread (standard deviation) for each score. The spread tells us how stable a model is and it matters as much as the average.

Table 5: Cross validation results on the training data (mean ± standard deviation)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
SVM (RBF) (tuned)	0.775 ± 0.029	0.751 ± 0.067	0.537 ± 0.064	0.623 ± 0.053	0.845 ± 0.025
Logistic Regression (tuned)	0.759 ± 0.040	0.649 ± 0.073	0.714 ± 0.054	0.675 ± 0.036	0.844 ± 0.026
Logistic Regression	0.783 ± 0.034	0.752 ± 0.074	0.574 ± 0.078	0.646 ± 0.064	0.843 ± 0.026
SVM (RBF)	0.763 ± 0.032	0.712 ± 0.077	0.561 ± 0.075	0.622 ± 0.050	0.832 ± 0.030
KNN	0.760 ± 0.039	0.705 ± 0.085	0.563 ± 0.061	0.621 ± 0.047	0.831 ± 0.039
Random Forest	0.756 ± 0.037	0.686 ± 0.071	0.572 ± 0.073	0.620 ± 0.057	0.829 ± 0.038
Gradient Boosting	0.757 ± 0.028	0.690 ± 0.058	0.567 ± 0.078	0.618 ± 0.051	0.829 ± 0.028
Naive Bayes	0.757 ± 0.032	0.681 ± 0.074	0.592 ± 0.049	0.630 ± 0.039	0.825 ± 0.035
Decision Tree	0.728 ± 0.040	0.633 ± 0.088	0.575 ± 0.114	0.592 ± 0.067	0.778 ± 0.042
Baseline (majority class)	0.651 ± 0.002	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.500 ± 0.000

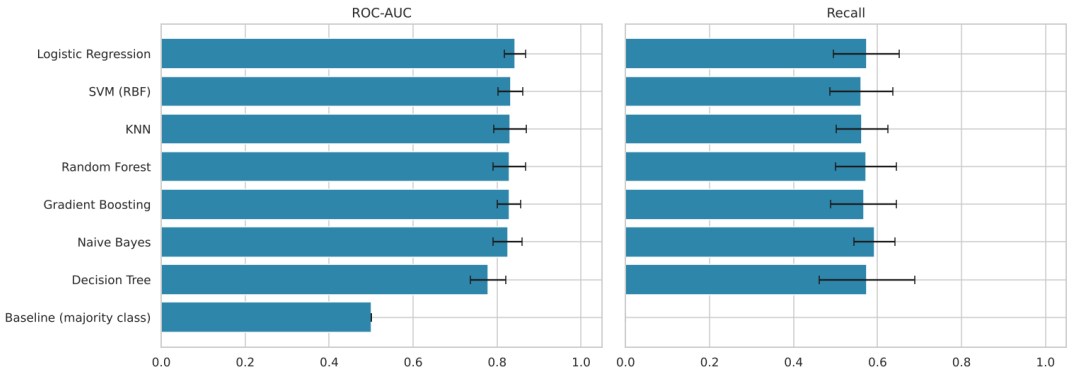


Figure 2: ROC-AUC and recall for each model, with the spread shown as error bars.

All seven models beat the baseline by a clear margin. Logistic regression came first among the untuned models (ROC-AUC 0.842). SVM, KNN, random forest, gradient boosting and naive Bayes were all within about 0.02 of it. The decision tree was clearly the weakest (0.778).

The spread is around 0.03 to 0.04 for most models. That is larger than the gaps between the top models, so the models at the top cannot honestly be called different from each other. The stability chart is in Appendix D.

6.2 Tuning the two strongest models

We tuned the two best models, Logistic Regression and SVM, with GridSearchCV (5 folds, scored on ROC-AUC, training data only). For Logistic Regression we searched C in 0.01, 0.1, 1 and 10 and class weights of none or balanced. For the SVM we searched C in 0.1, 1 and 10 and gamma in scale, 0.01 and 0.1.

Table 6: Tuning results

Model	Best settings found	Grid ROC-AUC
Logistic Regression	C = 0.1, class_weight = balanced	0.845
SVM (RBF)	C = 1, gamma = 0.01	0.846

The tuned models were then checked again with the same repeated cross validation as the others. They are the first two rows of Table 5.

6.3 Choosing the best model

The tuned SVM and the tuned logistic regression are almost identical on ROC-AUC (0.845 and 0.844). The gap of 0.001 is far smaller than the spread of about 0.026, so they are effectively tied.

We had set a rule before looking: if the top two are tied, pick the one with higher recall, because in this problem missing a sick patient is worse than a false alarm. The tuned logistic regression has recall of 0.714 against 0.538 for the tuned SVM. It is also the simpler and easier model to explain. So **tuned Logistic Regression** is our recommended model.

6.4 Final test on unseen patients

Now the test set was used, once. Table 7 compares the baseline and the chosen model at the usual threshold of 0.5.

Table 7: Test set results (154 patients, threshold 0.5)

	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline	0.649	0.000	0.000	0.000	0.50
Logistic Regression (tuned)	0.708	0.571	0.667	0.615	0.81

The model finds two out of every three patients with diabetes (recall 0.667) and gets a ROC-AUC of 0.81. That is a little lower than the 0.84 seen in cross validation. This is normal, because 154 patients is a small test set and one lucky or unlucky split moves the number and it is close to the 76% level reported by Smith et al.

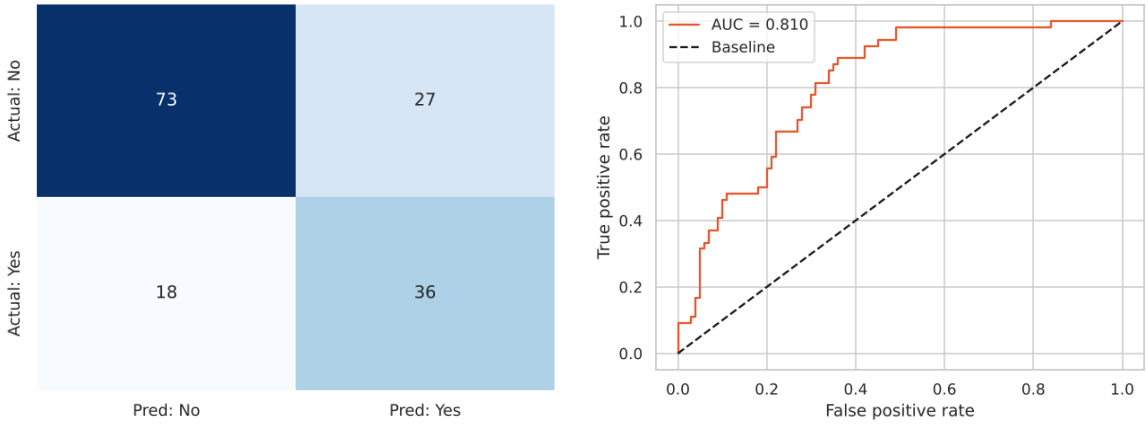


Figure 3: Confusion matrix (left) and ROC curve (right) on the test set, threshold 0.5.

Out of 154 patients, the model got 73 healthy and 36 diabetic patients right. It missed 18 patients with diabetes and raised a false alarm for 27 healthy patients.

6.5 Choosing a decision threshold

By default a patient is flagged when the predicted chance is 0.5 or more. But for screening it is better to catch more patients and accept more false alarms. We tested thresholds from 0.20 to 0.70 using predictions on the training data only, where each patient was scored by a model that had not seen them.

We used the F2 score, which counts recall twice as much as precision. It peaked at a threshold of 0.25.

Table 8: Threshold search on the training data (selected rows)

Threshold	Precision	Recall	F1	F2
0.20	0.454	0.986	0.622	0.799
0.25	0.489	0.963	0.649	0.807
0.30	0.500	0.911	0.646	0.783
0.40	0.564	0.822	0.669	0.753
0.50	0.650	0.720	0.683	0.704
0.60	0.724	0.626	0.672	0.644
0.70	0.767	0.523	0.622	0.559

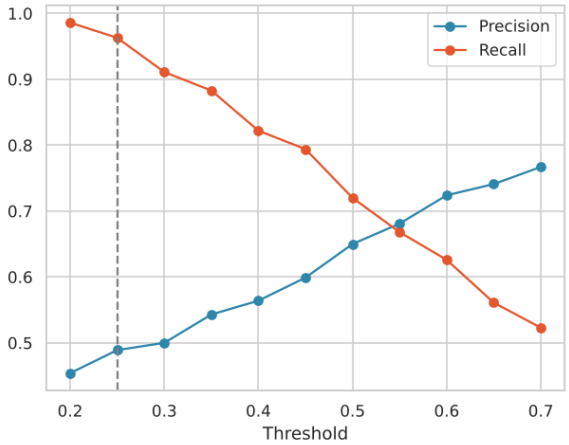


Figure 4: Precision and recall as the threshold changes. The dashed line marks 0.25.

Table 9: Test set at the two thresholds

	Accuracy	Precision	Recall	F1	Missed	False alarms
Threshold 0.50	0.708	0.571	0.667	0.615	18	27
Threshold 0.25	0.669	0.515	0.944	0.667	3	48

At 0.25 the model misses only 3 of the 54 patients with diabetes. The price is 48 false alarms among 100 healthy patients (up from 27). Whether that trade is fair depends on how cheap a follow-up test is. If a confirmatory test is cheap and the cost of a missed case is high, 0.25 is a sensible choice. If follow-up tests are scarce, something closer to 0.35 or 0.40 may suit better. That decision belongs to the clinic, not to the model.

6.6 Error analysis

We grouped the test patients into true positives, true negatives, false positives and false negatives (at threshold 0.5) and compared their average measurements.

Table 10: Average values for each type of result (measured values only)

Result	Patients	Glucose	BMI	Age	Insulin
True positive	36	152.7	36.6	39.2	262.6
False negative (missed)	18	114.6	31.4	30.5	136.6
False positive	27	142.0	36.5	38.0	277.8
True negative	73	100.2	29.2	28.1	124.7

There is a clear pattern. The patients the model missed look like healthy patients: they are younger (30.5 against 39.2 for caught cases) and their glucose is lower (114.6 against 152.7). The model leans heavily on glucose, so a diabetic patient with a normal-looking glucose reading is hard for it to catch.

The false alarms go the other way. Their glucose (142.0), BMI and age look much like those of true patients, so these are healthy women who simply look high risk on paper.

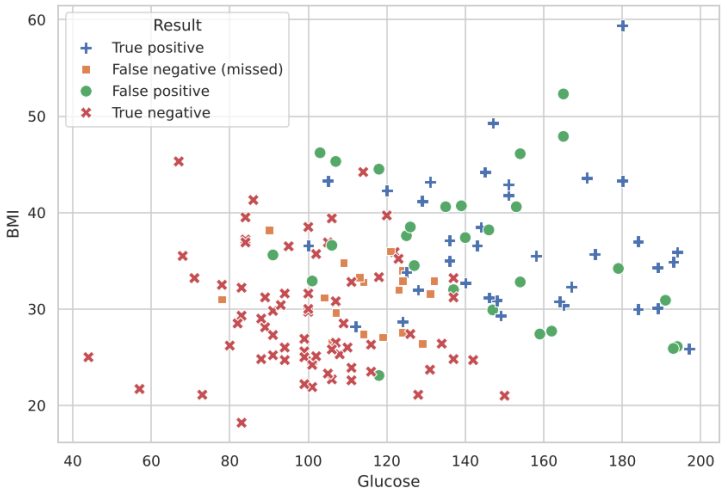


Figure 5: Glucose against BMI for each type of result.

6.7 Subgroup analysis

To check whether the model treats groups evenly, we used the out-of-fold predictions on the training data (614 patients), which gives more patients per group than the test set. Predictions use the 0.5 threshold.

Table 11: Performance by subgroup (training data, out-of-fold)

Group	Patients	Diabetic	Accuracy	Recall
Age 21–30	328	72	0.838	0.625
Age 31–40	130	58	0.715	0.707
Age 41–50	89	51	0.708	0.784
Age 51 and over	67	33	0.597	0.848
BMI under 25	85	7	0.941	0.714
BMI 25–29.9	141	30	0.745	0.400
BMI 30 and over	379	175	0.739	0.777
BMI not recorded	9	2	0.667	0.500
Pregnancies 0	85	31	0.753	0.613
Pregnancies 1–3	253	62	0.814	0.645
Pregnancies 4 or more	276	121	0.728	0.785

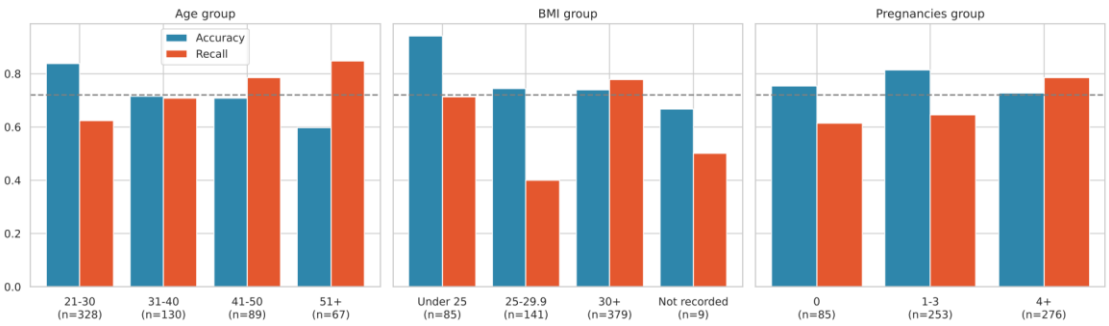


Figure 6: Accuracy and recall by group. The dashed line is the overall recall.

Performance is not even. Recall is best for the oldest women (0.848 for 51 and over) and for women with BMI of 30 or more (0.777). It is much weaker for women aged 21 to 30 (0.625) and, most clearly, for women with BMI between 25 and 29.9, where the model finds only 40% of the diabetic patients.

We must be careful with small groups. The under-25 BMI group has only 7 diabetic women and the age 51+ group has 67 patients, so those numbers can swing. But the BMI 25 to 29.9 gap (30 diabetic patients) is large enough to take seriously. In practice, a thin or average-weight woman with diabetes is the patient this model is most likely to miss.

6.8 Which features matter

We measured permutation importance on the test set, which shows how much the ROC-AUC drops when one feature is shuffled. Glucose dominates by a wide margin. BMI, Pregnancies and the family history score come next and Age adds a little. Insulin, SkinThickness and BloodPressure add almost nothing.

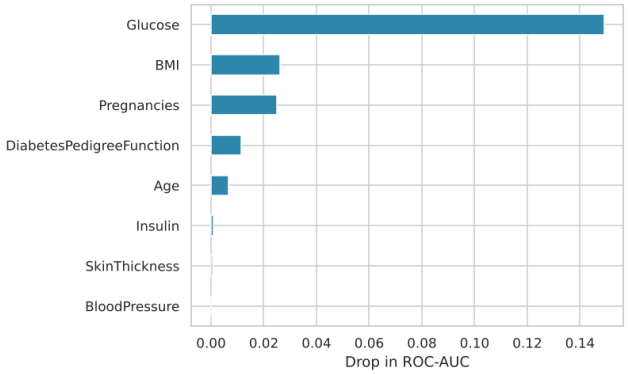


Figure 7: Permutation importance (drop in ROC-AUC when a feature is shuffled).

This agrees with the correlation results and it is a reminder of the earlier limitation. Because Glucose does most of the work, the model is only as useful as the glucose reading it receives.

7. Exploratory Component

We used PCA (principal component analysis) as an exploratory tool. It squeezes the eight features into new directions that capture as much variation as possible, so the data can be drawn on a flat page. We fitted it on the scaled training data only.

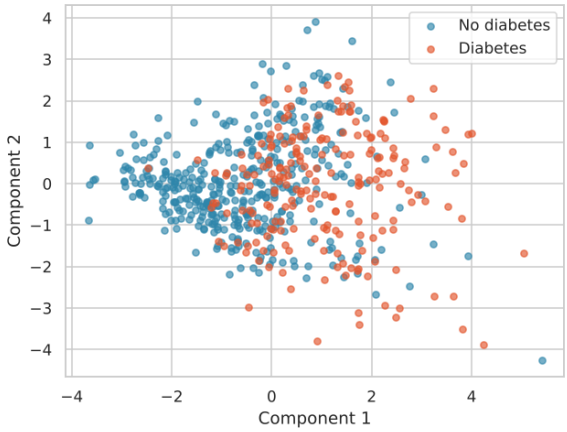


Figure 8: Training patients on the first two principal components.

The first two components explain 48.2% of the variation (29.4% and 18.8%) and it takes more than four components to pass 70%. The scatter plot shows that patients with and without diabetes overlap heavily, with only a soft drift between them.

This tells us two things. The classes are not neatly separated, which is why even the best model is far from perfect. And the information is spread over several features, so cutting the features down to two components before modelling would throw away useful signal. For that reason we kept PCA as an exploration only and did not use it inside the models. We did not use K-Means, because the task already has known labels and we could not see a clear job for clustering.

8. Responsible AI Assessment

8.1 Fairness

The model does not treat all groups equally. It misses more diabetic women who are young or who have a BMI between 25 and 29.9. It was also trained only on women of one community, so it knows nothing about men or about how other populations differ. Using it elsewhere without checking could give wrong results to groups it has never seen.

8.2 Privacy and consent

This is medical information about real people and the public dataset does not tell us what consent they gave for this kind of reuse. It comes from a single Indigenous community that has been studied for decades. We used only the public, anonymised file and we did not try to identify anyone. Any real deployment in Uganda would need patient consent, a clear data protection plan and approval from the right health authorities.

8.3 Transparency

Logistic regression is one of the easiest models to explain, because each feature has a weight that shows its direction and size. A clinician or a patient can be told, in plain words, that high glucose and high BMI pushed the score up. The permutation importance chart gives the same

story for the whole model. We think this was a good reason to prefer it over a model that is slightly more complex but not more accurate.

8.4 Human oversight

This model should never make a diagnosis on its own. It should only flag patients for a confirmatory laboratory test, with a clinician making the final call. Before using it for real we would want four things in place: a local validation on Ugandan patient data, a clinic-chosen threshold, regular checks of how many cases are being missed and a clear way for staff to override the flag.

Staff also need to understand the trade-off. At threshold 0.25 nearly half of the healthy women get flagged, so the follow-up system has to handle that load without causing fear or unnecessary cost for patients.

9. Reproducibility and Version Control

9.1 Repository and notebook

GitHub repository https://github.com/Alouzious/Diabetes-Risk-Prediction.git

Notebook
https://drive.google.com/file/d/1JGis4_X8qCUGR5WzGYJ1q1QVtaby9mwJ/view?usp=sharing

9.2 Deployment demonstration

The best pipeline was saved with joblib, loaded back and checked. The loaded model gave exactly the same predictions as the original (the check printed True). Then we gave it a new patient: a 45-year-old woman with 2 pregnancies, glucose 148, blood pressure 72, BMI 33.6 and a family history score of 0.5. Skin thickness and insulin were not measured, so they were left empty and the pipeline filled them in on its own.

The model gave her a **68.7%** chance of diabetes. That is above the 0.25 threshold, so the output was “Refer for lab test”.

10. Conclusion and Recommendations

The project built a complete, leakage-safe pipeline and compared seven models against a baseline. Tuned logistic regression came out as our recommendation. On the 154 held-out patients it reached a ROC-AUC of 0.81 and found 66.7% of diabetic patients at the default threshold, or 94.4% at a screening threshold of 0.25.

The real finding is how close the models were. Linear, kernel, tree and boosting models all landed within a narrow band, which suggests the limit is in the data and not in the algorithm.

10.1 Limitations

The data is small, from one community and only includes women. The diagnosis is tied to glucose, which is also a feature. The test set has just 154 patients, so the numbers carry some

luck. The threshold of 0.25 causes many false alarms and the model is weaker for some subgroups, especially women with a BMI of 25 to 29.9.

References

Kaufman, S., Rosset, S., Perlich, C. and Stitelman, O. (2012). Leakage in data mining: Formulation, detection and avoidance. ACM Transactions on Knowledge Discovery from Data, 6(4), Article 15.

Kavakiotis, I., Tsave, O., Salifoglou, A., Maglaveras, N., Vlahavas, I. and Chouvarda, I. (2017). Machine learning and data mining methods in diabetes research. Computational and Structural Biotechnology Journal, 15, 104–116. https://doi.org/10.1016/j.csbj.2016.12.005

Maniruzzaman, M., Kumar, N., Abedin, M. M., Islam, M. S., Suri, H. S., El-Baz, A. S. and Suri, J. S. (2017). Comparative approaches for classification of diabetes mellitus data: Machine learning paradigm. Computer Methods and Programs in Biomedicine, 152, 23–34.

National Institute of Diabetes and Digestive and Kidney Diseases. Pima Indians Diabetes Database. Donated by V. Sigillito to the UCI Machine Learning Repository, 1990. Also available on Kaggle: https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database

Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

Smith, J. W., Everhart, J. E., Dickson, W. C., Knowler, W. C. and Johannes, R. S. (1988). Using the ADAP learning algorithm to forecast the onset of diabetes mellitus. Proceedings of the Symposium on Computer Applications and Medical Care, 261–265. IEEE Computer Society Press.

Zou, Q., Qu, K., Luo, Y., Yin, D., Ju, Y. and Tang, H. (2018). Predicting diabetes mellitus with machine learning techniques. Frontiers in Genetics, 9, 515. https://doi.org/10.3389/fgene.2018.00515

Appendices

Appendix A: Histogram of each feature

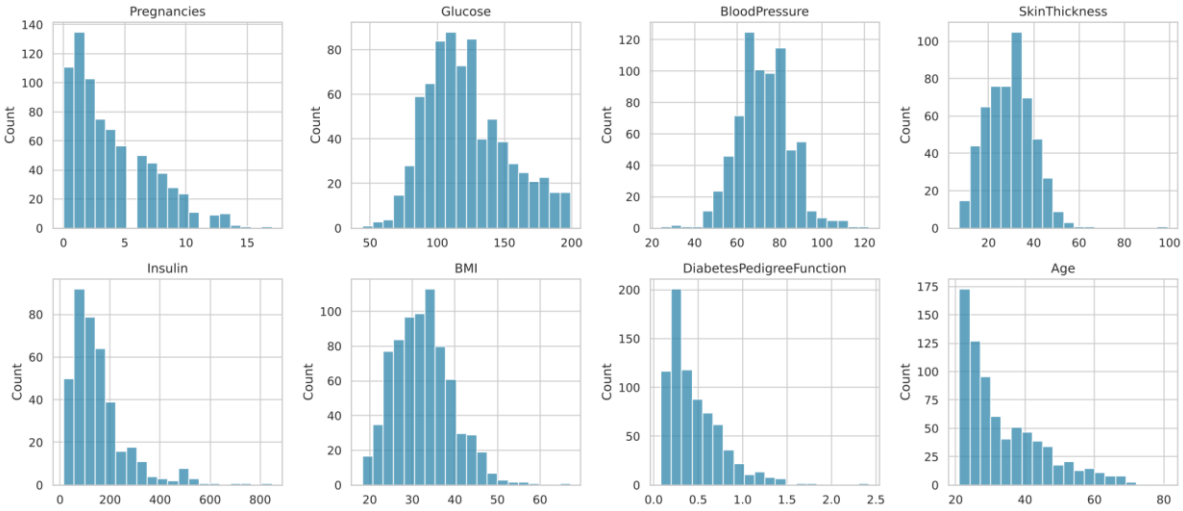


Figure A1: Distribution of each feature (missing values left out).

Appendix B: Boxplots by outcome

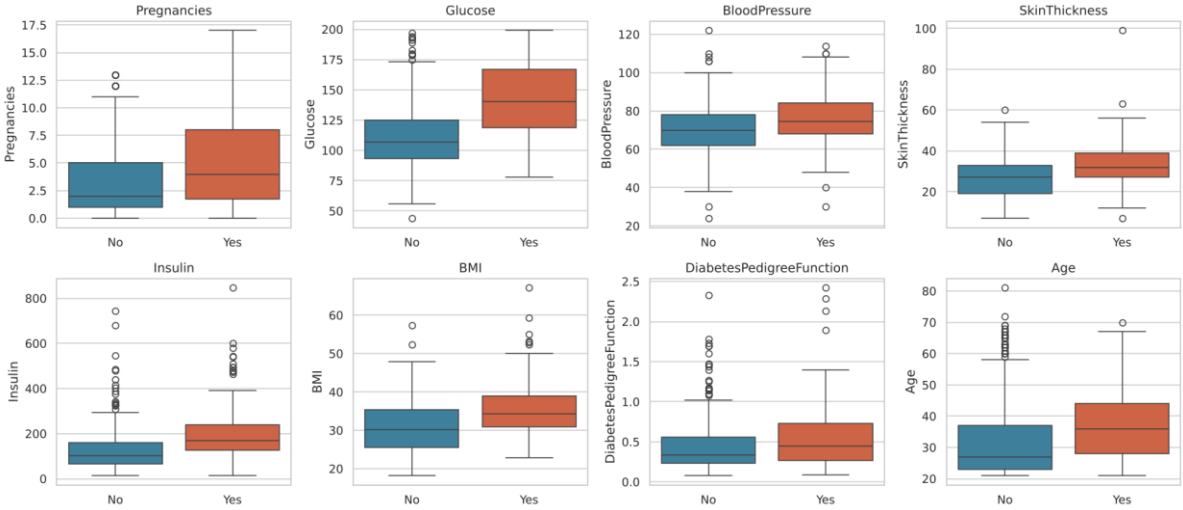


Figure A2: Each feature split by outcome (No = no diabetes, Yes = diabetes).

Appendix C: Correlation heatmap

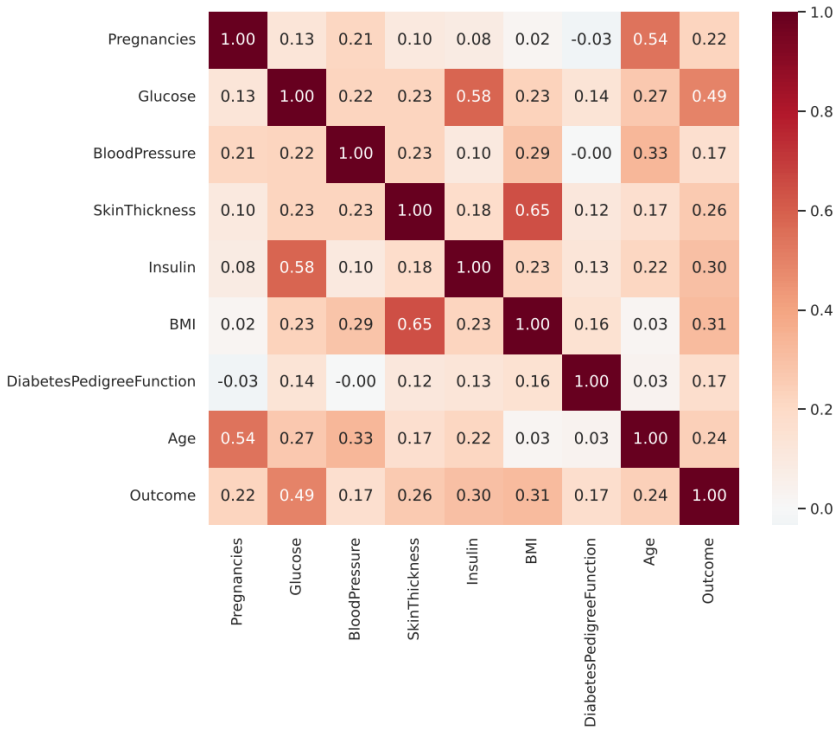


Figure A3: Correlation between all features and the outcome.

Appendix D: Stability of each model

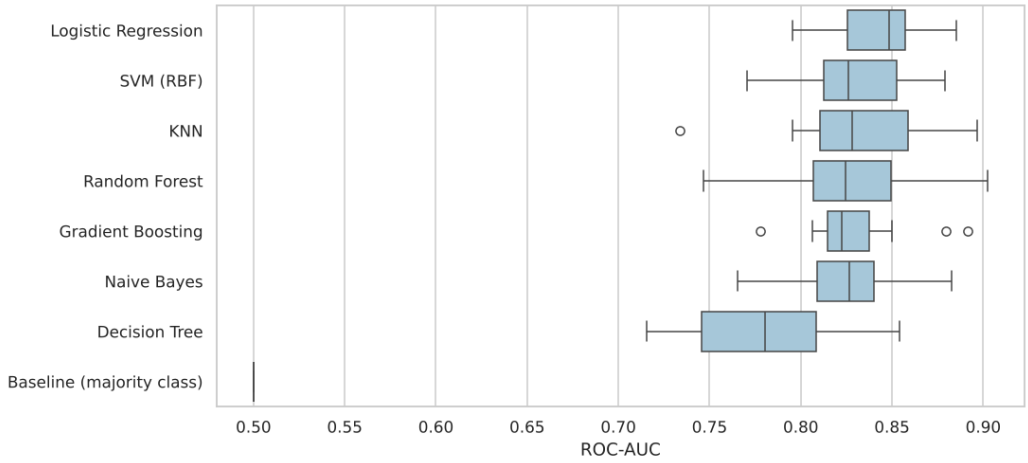


Figure A4: ROC-AUC across the 15 cross validation scores for each model.